

Calculus, Trig & Beyond

A rigorous, self-contained reference covering derivatives, integrals, logarithms, trig identities, implicit differentiation, curve sketching, volumes of revolution, and advanced integration — with full proofs, exhaustive worked examples, and common trap warnings.

DERIVATIVES

INTEGRALS

LOGARITHMS

TRIG IDENTITIES

QUADRANTS

IMPLICIT DIFF.

CURVE SKETCHING

VOLUMES

U-SUB

IBP

PARTIAL FRACTIONS

§ 01 General Derivatives

Core differentiation theory, formal definitions, rule proofs, and graded examples.

1.1 — THE LIMIT DEFINITION OF THE DERIVATIVE

DEFINITION — THE DERIVATIVE

The derivative of a function f at a point x is defined as the limit of the difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists. Geometrically, $f'(x)$ is the slope of the tangent line to the graph of f at the point $(x, f(x))$. Equivalently:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

A function is *differentiable* at x if and only if this limit exists and is finite. Differentiability implies continuity, but continuity does not imply differentiability.

ALTERNATIVE NOTATIONS

$$f'(x) = \frac{dy}{dx} = \frac{d}{dx}[f(x)] = Df(x) = \dot{y}$$

Higher-order derivatives: $f''(x) = \frac{d^2y}{dx^2}$, and $f^{(n)}(x) = \frac{d^ny}{dx^n}$.

1.2 — CORE DIFFERENTIATION RULES

CONSTANT RULE

$$\frac{d}{dx}[c] = 0$$

Any constant has zero rate of change.

CONSTANT MULTIPLE RULE

$$\frac{d}{dx}[c f(x)] = c f'(x)$$

SUM / DIFFERENCE RULE

$$\frac{d}{dx}[f \pm g] = f' \pm g'$$

POWER RULE

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

Valid for all real n .

PRODUCT RULE

$$\frac{d}{dx}[fg] = f'g + fg'$$

QUOTIENT RULE

$$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{f'g - fg'}{g^2}$$

"Lo d-Hi minus Hi d-Lo over Lo squared."

CHAIN RULE – COMPOSITE FUNCTIONS

If $y = f(g(x))$, then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

In Leibniz notation: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Apply from outermost function inward — "outside in."

1.3 — DERIVATIONS & PROOFS

PROOF — POWER RULE VIA THE BINOMIAL THEOREM

We prove $\frac{d}{dx}[x^n] = nx^{n-1}$ for positive integers n .

$$\frac{d}{dx}[x^n] = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

Expand $(x+h)^n$ via the Binomial Theorem:

$$(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots + h^n$$

Subtract x^n , divide by h , and take $h \rightarrow 0$. All terms with h vanish:

$$\lim_{h \rightarrow 0} \left[nx^{n-1} + \binom{n}{2}x^{n-2}h + \dots \right] = nx^{n-1}$$

■ Q.E.D.

PROOF — PRODUCT RULE FROM THE LIMIT DEFINITION

We prove $\frac{d}{dx}[fg] = f'g + fg'$. Add and subtract $f(x+h)g(x)$ in the numerator:

$$\begin{aligned} \frac{d}{dx}[fg] &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[f(x+h) \cdot \frac{g(x+h) - g(x)}{h} \right] + \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \cdot g(x) \right] = f(x)g'(x) + f'(x)g(x) \end{aligned}$$

■ Q.E.D.

DERIVATION — QUOTIENT RULE FROM THE PRODUCT RULE

Write $\frac{f}{g} = f \cdot g^{-1}$. Apply Product Rule and Chain Rule:

$$\frac{d}{dx}\left[\frac{f}{g}\right] = f' \cdot g^{-1} + f \cdot (-g^{-2})g' = \frac{f'}{g} - \frac{fg'}{g^2} = \frac{f'g - fg'}{g^2}$$

■ Q.E.D.

1.4 — WORKED EXAMPLES

EASY Differentiate $f(x) = 4x^3 - 7x^2 + 2x - 9$

POWER RULE TERM BY TERM

$$f'(x) = 12x^2 - 14x + 2$$

MEDIUM Differentiate $g(x) = x^2e^{3x}$

PRODUCT RULE: $U = x^2$, $V = e^{3x}$

$$g'(x) = 2x \cdot e^{3x} + x^2 \cdot 3e^{3x}$$

$$g'(x) = xe^{3x}(2 + 3x)$$

HARD Differentiate $h(x) = \frac{(x^2 + 1)^4}{(3x - 2)^3}$

QUOTIENT RULE WITH CHAIN RULE IN NUMERATOR AND DENOMINATOR

$$f = (x^2 + 1)^4, \quad f' = 8x(x^2 + 1)^3$$

$$g = (3x - 2)^3, \quad g' = 9(3x - 2)^2$$

QUOTIENT RULE THEN FACTOR $(x^2 + 1)^3(3x - 2)^2$

$$h'(x) = \frac{(x^2 + 1)^3(3x - 2)^2[8x(3x - 2) - 9(x^2 + 1)]}{(3x - 2)^6}$$

$$\text{Bracket: } 24x^2 - 16x - 9x^2 - 9 = 15x^2 - 16x - 9$$

$$h'(x) = \frac{(x^2 + 1)^3(15x^2 - 16x - 9)}{(3x - 2)^4}$$

⚠ COMMON TRAPS – DERIVATIVES

Trap 1 — Forgetting the Chain Rule: $\frac{d}{dx}[\sin(x^2)] \neq \cos(x^2)$. Must multiply by the inner derivative: $= 2x \cos(x^2)$.

Trap 2 — Power rule on exponentials: $\frac{d}{dx}[e^x] \neq xe^{x-1}$. The power rule does NOT apply when the variable is in the exponent.

Trap 3 — Quotient rule sign: It's $f'g - fg'$, NOT $fg' - f'g$. The sign order matters.

Trap 4 — $\frac{d}{dx}[f(x)^{g(x)}]$: Neither the power rule nor the exponential rule apply directly. Use logarithmic differentiation.

5 02 General Integrals

Indefinite and definite integration, the Fundamental Theorem of Calculus, and geometric interpretation.

2.1 — DEFINITION & THE FUNDAMENTAL THEOREM OF CALCULUS

INDEFINITE INTEGRAL

$$\int f(x) \, dx = F(x) + C \quad \text{where } F'(x) = f(x)$$

FTC PART 1

If f is continuous on $[a, b]$:

$$\frac{d}{dx} \int_a^x f(t) \, dt = f(x)$$

FTC PART 2

$$\int_a^b f(x) \, dx = F(b) - F(a) = \left[F(x) \right]_a^b$$

GENERALISED FTC1 (LEIBNIZ RULE)

$$\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) \, dt = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x)$$

2.2 — POWER RULE, LINEARITY, & CORE FORMULAS

POWER RULE ($N \neq -1$)

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

THE $N = -1$ EXCEPTION

$$\int \frac{1}{x} \, dx = \ln |x| + C$$

Power rule gives $x^0/0$ — undefined. The correct antiderivative is $\ln |x|$.

CONSTANT MULTIPLE

$$\int c f(x) \, dx = c \int f(x) \, dx$$

SUM / DIFFERENCE

$$\int [f \pm g] \, dx = \int f \, dx \pm \int g \, dx$$

2.3 — THE CONSTANT OF INTEGRATION & GEOMETRY

WHY $+C$ EXISTS

Differentiation destroys constant information (derivative of any constant is 0). The antiderivative is an entire *family* of functions, each differing by a vertical shift. The constant C is pinned down by initial conditions or boundary values.

GEOMETRIC INTERPRETATION — SIGNED AREA

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x, \quad \Delta x = \frac{b-a}{n}$$

Area above the x -axis is positive; below is negative. For unsigned area use $\int_a^b |f(x)| \, dx$.

2.4 — WORKED EXAMPLES

EASY Evaluate $\int (5x^4 - 3x^2 + 2x - 7) \, dx$

$$x^5 - x^3 + x^2 - 7x + C$$

MEDIUM Evaluate $\int_1^4 \frac{3x^2 + \sqrt{x} - 1}{x} dx$

DIVIDE BY x , THEN INTEGRATE

$$\int_1^4 \left(3x + x^{-1/2} - x^{-1}\right) dx = \left[\frac{3x^2}{2} + 2\sqrt{x} - \ln x\right]_1^4$$

At $x = 4$: $24 + 4 - \ln 4$. At $x = 1$: $\frac{3}{2} + 2 - 0 = \frac{7}{2}$.

$$\frac{49}{2} - \ln 4$$

HARD Evaluate $\int \frac{x^3 - 2x^2 + 5}{x^2} dx$

SPLIT: $\int (X - 2 + 5X^{-2}) DX$

$$\frac{x^2}{2} - 2x - \frac{5}{x} + C$$

VERIFY BY DIFFERENTIATING:

$$\frac{d}{dx}[\dots] = x - 2 + \frac{5}{x^2} = \frac{x^3 - 2x^2 + 5}{x^2} \checkmark$$

⚠ COMMON TRAPS – INTEGRALS

Trap 1 — Forgetting $+C$ in indefinite integrals: Always required unless bounds are given.

Trap 2 — $n = -1$ case: $\int x^{-1} dx = \ln|x| + C$, NOT $\frac{x^0}{0}$.

Trap 3 — Absolute value in $\ln$: $\int \frac{1}{x} dx = \ln|x| + C$, not $\ln(x) + C$. The domain includes negative x .

Trap 4 — Changing bounds in definite integrals with u-sub: You must transform the bounds using the substitution, OR back-substitute before evaluating.

5
03

Logarithms & Exponentials

Definitions, properties, Euler's number, derivatives, integrals, proofs, and exhaustive worked examples.

3.1

— LOGARITHM DEFINITIONS

LOGARITHM — DEFINITION

$$\log_a x = y \iff a^y = x \text{ where } a > 0, \ a \neq 1, \ x > 0.$$

The **natural logarithm** uses base e :

$$\ln x = \int_1^x \frac{1}{t} dt, \qquad x > 0$$

This integral definition is the most rigorous — it constructs $\ln x$ without presupposing e .

Inverse relationship: $e^{\ln x} = x$ ($x > 0$) and $\ln(e^x) = x$ ($x \in \mathbb{R}$).

3.2 — CORE PROPERTIES OF LOGARITHMS

PRODUCT RULE

$$\log_a(MN) = \log_a M + \log_a N$$

QUOTIENT RULE

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

POWER RULE

$$\log_a(M^p) = p \log_a M$$

CHANGE OF BASE

$$\log_a x = \frac{\ln x}{\ln a}$$

CANCELLATION IDENTITIES

$$a^{\log_a x} = x \quad \log_a(a^x) = x$$

SPECIAL VALUES

$$\ln 1 = 0, \quad \ln e = 1, \quad \log_a 1 = 0, \quad \log_a a = 1$$

3.3 — EULER'S NUMBER e

EULER'S NUMBER — LIMIT DEFINITION

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828\,18284\dots$$

Also: $e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \dots$

The number e is transcendental — not a root of any nonzero polynomial with rational coefficients. Its defining calculus property:

$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$, meaning e^x has slope 1 at the origin.

3.4 — DERIVATIVES OF EXPONENTIAL & LOGARITHMIC FUNCTIONS

FUNCTION	DERIVATIVE	CHAIN RULE FORM
e^x	e^x	$e^u \cdot u'$
a^x	$a^x \ln a$	$a^u \ln a \cdot u'$
$\ln x$	$\frac{1}{x}$	$\frac{u'}{u}$
$\ln x $	$\frac{1}{x}$	$\frac{u'}{u}$
$\log_a x$	$\frac{1}{x \ln a}$	$\frac{u'}{u \ln a}$

3.5 — INTEGRALS OF EXPONENTIAL & LOGARITHMIC FUNCTIONS

INTEGRAL	RESULT
$\int e^x \, dx$	$e^x + C$
$\int e^{ax} \, dx$	$\frac{e^{ax}}{a} + C$

INTEGRAL	RESULT
$\int a^x \, dx$	$\frac{a^x}{\ln a} + C$
$\int \frac{1}{x} \, dx$	$\ln x + C$
$\int \ln x \, dx$	$x \ln x - x + C$
$\int \log_a x \, dx$	$\frac{x \ln x - x}{\ln a} + C$
$\int \frac{f'(x)}{f(x)} \, dx$	$\ln f(x) + C$

3.6 — FULL PROOFS

PROOF — DERIVATIVE OF $\ln x$ VIA IMPLICIT DIFFERENTIATION

Let $y = \ln x$, so $e^y = x$. Differentiate both sides w.r.t. x :

$$e^y \cdot \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

■ Q.E.D.

PROOF — DERIVATIVE OF e^x

$$\frac{d}{dx}[e^x] = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x$$

■ Q.E.D.

DERIVATION — DERIVATIVE OF A^x

Write $a^x = e^{x \ln a}$ and apply Chain Rule:

$$\frac{d}{dx}[a^x] = e^{x \ln a} \cdot \ln a = a^x \ln a$$

■

DERIVATION — $\int \ln x \, dx$ BY INTEGRATION BY PARTS

Write as $\int \ln x \cdot 1 \, dx$. Set $u = \ln x$, $dv = dx$, so $du = \frac{1}{x} dx$, $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - x + C$$

Verify: $\frac{d}{dx}[x \ln x - x] = \ln x + 1 - 1 = \ln x \checkmark$

$$\int \ln x \, dx = x \ln x - x + C$$

■ Q.E.D.

3.7 — WORKED EXAMPLES

EASY Differentiate $f(x) = e^{3x^2-5}$

CHAIN RULE: OUTER e^u , INNER $u = 3x^2 - 5$, $u' = 6x$

$f'(x) = 6x e^{3x^2 - 5}$

MEDIUM

Differentiate $g(x) = \ln\left(\frac{(x^2 + 1)^3}{x^4 + 5}\right)$

EXPAND WITH LOG PROPERTIES FIRST

$$g(x) = 3 \ln(x^2 + 1) - \ln(x^4 + 5)$$

$g'(x) = \frac{6x}{x^2 + 1} - \frac{4x^3}{x^4 + 5}$

HARD

Logarithmic differentiation: $\frac{d}{dx}[x^{\sin x}]$

LET $Y = X^{\sin X}$. TAKE $\ln Y = \sin X \cdot \ln X$, THEN DIFFERENTIATE IMPLICITLY

$$\frac{1}{y} \frac{dy}{dx} = \cos x \ln x + \frac{\sin x}{x}$$

$\frac{dy}{dx} = x^{\sin x} \left(\ln x \cos x + \frac{\sin x}{x} \right)$

VERY HARD

Logarithmic differentiation of the tower $y = x^{(x^x)}$

SUB-PROBLEM: $\frac{D}{DX}[X^X]$. LET $W = X^X \Rightarrow \ln W = X \ln X$

$$\frac{dw}{dx} = x^x (\ln x + 1)$$

MAIN: $\ln Y = X^X \ln X$. PRODUCT RULE ON THE RIGHT:

$$\frac{1}{y} \frac{dy}{dx} = x^x (\ln x + 1) \ln x + x^x \cdot \frac{1}{x} = x^x \left[\ln^2 x + \ln x + \frac{1}{x} \right]$$

$\frac{dy}{dx} = x^{(x^x)} \cdot x^x \left(\ln^2 x + \ln x + \frac{1}{x} \right)$

⚠ COMMON TRAPS – LOGS & EXPONENTIALS

Trap 1 — $\ln(a + b) \neq \ln a + \ln b$: The product rule for logs only applies to $\ln(ab)$.

Trap 2 — $\ln(a^b) = b \ln a$ **only for** $a > 0$: Check domain first.

Trap 3 — **Forgetting absolute value**: $\int \frac{1}{x} dx = \ln|x| + C$. Without $|\cdot|$ the formula is only valid for $x > 0$.

Trap 4 — **Differentiating $\ln(f(x))$ without Chain Rule**: $\frac{d}{dx}[\ln(x^2 + 1)] = \frac{2x}{x^2 + 1}$, not $\frac{1}{x^2 + 1}$.

FORMULA & METHODOLOGY

If $u = g(x)$, then $du = g'(x) \, dx$, and:

$$\int f(g(x)) \, g'(x) \, dx = \int f(u) \, du$$

For definite integrals: $\int_a^b f(g(x))g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du$

Method: (1) Choose u as the inner function. (2) Compute du . (3) Substitute and eliminate all x . (4) Integrate. (5) Back-substitute.

MEDIUM Evaluate $\int x^2 e^{x^3+1} \, dx$

Let $u = x^3 + 1 \Rightarrow du = 3x^2 \, dx \Rightarrow x^2 \, dx = du/3$.

$$\int e^u \cdot \frac{du}{3} = \frac{1}{3}e^u + C = \boxed{\frac{1}{3}e^{x^3+1} + C}$$

VERY HARD Evaluate $\int_e^{e^2} \frac{\ln(\ln x)}{x \ln x} \, dx$

FIRST: $U = \ln X \Rightarrow DU = DX/X$. BOUNDS: $E \rightarrow 1, E^2 \rightarrow 2$

$$\int_1^2 \frac{\ln u}{u} \, du$$

SECOND: $V = \ln U \Rightarrow DV = DU/U$. BOUNDS: $1 \rightarrow 0, 2 \rightarrow \ln 2$

$$\int_0^{\ln 2} v \, dv = \left[\frac{v^2}{2} \right]_0^{\ln 2} = \boxed{\frac{(\ln 2)^2}{2}}$$

FORMULA — INTEGRATION BY PARTS

Derived from the Product Rule:

$$\int u \, dv = uv - \int v \, du$$

$$\int_a^b u \, dv = \left[uv \right]_a^b - \int_a^b v \, du$$

LIATE RULE — CHOOSING U

Choose u as the earliest in the LIATE hierarchy.

L	I	A	T	E
LOGARITHMS	INVERSE TRIG	ALGEBRAIC	TRIG	EXPONENTIAL

MEDIUM Evaluate $\int x^2 \ln x \, dx$

LIATE: L beats A. $u = \ln x, dv = x^2 dx \Rightarrow v = x^3/3$.

$$\int x^2 \ln x \, dx = \frac{x^3}{3} \ln x - \int \frac{x^3}{3} \cdot \frac{1}{x} \, dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C$$

$$= \frac{x^3}{9} (3 \ln x - 1) + C$$

VERY HARD

Evaluate $\int e^x \sin x \, dx$ (cyclic IBP)

Let $I = \int e^x \sin x \, dx$. $u = \sin x$, $dv = e^x dx$:

$$I = e^x \sin x - \int e^x \cos x \, dx$$

Apply IBP again to the new integral ($u = \cos x$):

$$\int e^x \cos x \, dx = e^x \cos x + I$$

Substitute back: $I = e^x \sin x - e^x \cos x - I \Rightarrow 2I = e^x (\sin x - \cos x)$

$$\int e^x \sin x \, dx = \frac{e^x (\sin x - \cos x)}{2} + C$$

4.3 — PARTIAL FRACTION DECOMPOSITION

METHOD OVERVIEW

Decomposes $\frac{P(x)}{Q(x)}$ (with $\deg P < \deg Q$) into simpler fractions.

Step 0: If $\deg P \geq \deg Q$, perform polynomial long division first.

Step 1: Factor $Q(x)$ over $\mathbb{R}$.

Step 2: Write decomposition templates.

Step 3: Clear denominators and solve for constants.

Step 4: Integrate each term.

Templates:

- Distinct linear $(ax + b)$: $\frac{A}{ax + b}$
- Repeated linear $(ax + b)^k$: $\frac{A_1}{ax + b} + \dots + \frac{A_k}{(ax + b)^k}$
- Irreducible quadratic $(ax^2 + bx + c)$: $\frac{Ax + B}{ax^2 + bx + c}$

VERY HARD

Evaluate $\int \frac{3x^3 + 7x^2 + 2x + 3}{x^2(x^2 + x + 1)} \, dx$

x^2 IS A REPEATED LINEAR FACTOR; x^2+x+1 HAS $\Delta = -3 < 0$, SO IRREDUCIBLE QUADRATIC

$$\frac{3x^3 + 7x^2 + 2x + 3}{x^2(x^2 + x + 1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + x + 1}$$

Expanding and equating coefficients:

$$B = 3, \quad A = -1, \quad D = 5, \quad C = 4$$

INTEGRATE EACH TERM. FOR THE QUADRATIC: WRITE $4X+5 = 2(2X+1) + 3$

$$\int \frac{4x + 5}{x^2 + x + 1} \, dx = 2 \ln(x^2 + x + 1) + 2\sqrt{3} \arctan \frac{2x + 1}{\sqrt{3}}$$

$$-\ln|x| - \frac{3}{x} + 2 \ln(x^2 + x + 1) + 2\sqrt{3} \arctan \frac{2x + 1}{\sqrt{3}} + C$$

⚠ COMMON TRAPS — INTEGRATION TECHNIQUES

Trap 1 — IBP with cyclic integrals: When the original integral reappears, collect it to one side and solve algebraically. Do NOT apply IBP a third time.

Trap 2 — U-sub bounds: For definite integrals, either change bounds using $u = g(x)$, or back-substitute before evaluating. Mixing both leads to errors.

Trap 3 — Partial fractions without long division: If the degree of the numerator $\geq$ degree of denominator, divide first.

Trap 4 — Irreducible quadratic numerator form: For $\frac{Ax + B}{ax^2 + bx + c}$, the numerator must be linear, not constant.

5 05 Trigonometry

Fundamental identities, quadrant analysis (CAST rule), compound & double angle formulas, trig derivatives, integrals, and worked examples.

5.1 — FUNDAMENTAL TRIGONOMETRIC IDENTITIES

PYTHAGOREAN IDENTITY (PRIMARY)

$\sin^2 \theta + \cos^2 \theta = 1$

Derived from the unit circle $x^2 + y^2 = 1$.

PYTHAGOREAN IDENTITY (TAN)

$1 + \tan^2 \theta = \sec^2 \theta$

Divide the primary identity by $\cos^2 \theta$.

PYTHAGOREAN IDENTITY (COT)

$1 + \cot^2 \theta = \csc^2 \theta$

Divide the primary identity by $\sin^2 \theta$.

RECIPROCAL IDENTITIES

$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}$

QUOTIENT IDENTITIES

$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$

EVEN / ODD IDENTITIES

$\cos(-\theta) = \cos \theta \quad (\text{even})$

$\sin(-\theta) = -\sin \theta \quad (\text{odd})$

$\tan(-\theta) = -\tan \theta \quad (\text{odd})$

CO-FUNCTION IDENTITIES

$\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$

$\tan \theta = \cot\left(\frac{\pi}{2} - \theta\right)$

$\sec \theta = \csc\left(\frac{\pi}{2} - \theta\right)$

KEY ANGLE VALUES (UNIT CIRCLE)

ANGLE θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$	2π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.	0	undef.	0

5.2 — QUADRANTS & THE CAST RULE

THE CAST RULE (SIGNS OF TRIG FUNCTIONS BY QUADRANT)

Starting from Quadrant IV and going anticlockwise, the mnemonic **CAST** tells which function is *positive*:
C = Cosine positive (Q4), **A** = All positive (Q1), **S** = Sine positive (Q2), **T** = Tan positive (Q3).

Quadrant II (90° – 180°)

$\sin \theta > 0$, $\cos \theta < 0$, $\tan \theta < 0$
Only **Sine** (and cosecant) is positive.
Reference angle: $\pi - \theta$

Quadrant I (0° – 90°)

$\sin \theta > 0$, $\cos \theta > 0$, $\tan \theta > 0$
All functions are positive.
Reference angle: θ

Quadrant III (180° – 270°)

$\sin \theta < 0$, $\cos \theta < 0$, $\tan \theta > 0$
Only **Tangent** (and cotangent) is positive.
Reference angle: $\theta - \pi$

Quadrant IV (270° – 360°)

$\sin \theta < 0$, $\cos \theta > 0$, $\tan \theta < 0$
Only **Cosine** (and secant) is positive.
Reference angle: $2\pi - \theta$

USING REFERENCE ANGLES

To evaluate $\sin(5\pi/6)$: the angle is in Q2, so sine is positive. Reference angle = $\pi - 5\pi/6 = \pi/6$. Therefore:

$$\sin\left(\frac{5\pi}{6}\right) = +\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

To evaluate $\cos(7\pi/4)$: Q4, cosine positive. Reference angle = $2\pi - 7\pi/4 = \pi/4$:

$$\cos\left(\frac{7\pi}{4}\right) = +\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

To evaluate $\tan(4\pi/3)$: Q3, tangent positive. Reference angle = $4\pi/3 - \pi = \pi/3$:

$$\tan\left(\frac{4\pi}{3}\right) = +\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$$

5.3 — COMPOUND ANGLE, DOUBLE ANGLE & HALF ANGLE FORMULAS

COMPOUND — SINE

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

COMPOUND — COSINE

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

COMPOUND — TANGENT

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

DOUBLE ANGLE — SINE

$$\sin 2A = 2 \sin A \cos A$$

DOUBLE ANGLE — COSINE (3 FORMS)

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 \\ &= 1 - 2 \sin^2 A\end{aligned}$$

DOUBLE ANGLE — TANGENT

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

HALF-ANGLE FORMULAS

$\sin^2 A = \frac{1 - \cos 2A}{2}$

$\cos^2 A = \frac{1 + \cos 2A}{2}$

$\tan^2 A = \frac{1 - \cos 2A}{1 + \cos 2A}$

The half-angle formulas are essential for integrating even powers of sin and cos.

PRODUCT-TO-SUM FORMULAS

$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$

$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$

$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$

5.4 — DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

FUNCTION	DERIVATIVE	CHAIN RULE FORM ($U = U(x)$)
$\sin x$	$\cos x$	$\cos(u) \cdot u'$
$\cos x$	$-\sin x$	$-\sin(u) \cdot u'$
$\tan x$	$\sec^2 x$	$\sec^2(u) \cdot u'$
$\cot x$	$-\csc^2 x$	$-\csc^2(u) \cdot u'$
$\sec x$	$\sec x \tan x$	$\sec(u) \tan(u) \cdot u'$
$\csc x$	$-\csc x \cot x$	$-\csc(u) \cot(u) \cdot u'$
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{u'}{\sqrt{1-u^2}}$
$\arccos x$	$\frac{-1}{\sqrt{1-x^2}}$	$\frac{-u'}{\sqrt{1-u^2}}$
$\arctan x$	$\frac{1}{1+x^2}$	$\frac{u'}{1+u^2}$

PROOF — DERIVATIVE OF $\sin x$ FROM FIRST PRINCIPLES

Using the compound angle formula $\sin(x + h) = \sin x \cos h + \cos x \sin h$:

$$\begin{aligned} \frac{d}{dx} [\sin x] &= \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h} = \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \sin x \underbrace{\lim_{h \rightarrow 0} \frac{\cos h - 1}{h}}_{=0} + \cos x \underbrace{\lim_{h \rightarrow 0} \frac{\sin h}{h}}_{=1} = \cos x \end{aligned}$$

Two key limits used: $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ and $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$.

■ Q.E.D.

PROOF — DERIVATIVE OF $\cos x$

Using $\cos(x + h) = \cos x \cos h - \sin x \sin h$:

$$\frac{d}{dx} [\cos x] = \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \cos x \cdot 0 - \sin x \cdot 1 = -\sin x$$

■ Q.E.D.

PROOF — DERIVATIVE OF $\tan x$ VIA QUOTIENT RULE

$$\frac{d}{dx}[\tan x] = \frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

■ Q.E.D.

PROOF — DERIVATIVE OF $\arctan x$

Let $y = \arctan x$, so $\tan y = x$. Differentiate implicitly:

$$\sec^2 y \cdot \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

■ Q.E.D.

5.5 — INTEGRALS OF TRIGONOMETRIC FUNCTIONS

Integral	Result	Notes
$\int \sin x \, dx$	$-\cos x + C$	
$\int \cos x \, dx$	$\sin x + C$	
$\int \tan x \, dx$	$\ln \sec x + C$	$= -\ln \cos x + C$
$\int \cot x \, dx$	$\ln \sin x + C$	
$\int \sec x \, dx$	$\ln \sec x + \tan x + C$	Non-obvious; derived by multiplying by $\frac{\sec x + \tan x}{\sec x + \tan x}$
$\int \csc x \, dx$	$\ln \csc x - \cot x + C$	
$\int \sec^2 x \, dx$	$\tan x + C$	
$\int \csc^2 x \, dx$	$-\cot x + C$	
$\int \sec x \tan x \, dx$	$\sec x + C$	
$\int \csc x \cot x \, dx$	$-\csc x + C$	
$\int \frac{1}{\sqrt{1-x^2}} \, dx$	$\arcsin x + C$	$ x < 1$
$\int \frac{1}{1+x^2} \, dx$	$\arctan x + C$	
$\int \sin^2 x \, dx$	$\frac{x}{2} - \frac{\sin 2x}{4} + C$	Use half-angle: $\sin^2 x = \frac{1-\cos 2x}{2}$
$\int \cos^2 x \, dx$	$\frac{x}{2} + \frac{\sin 2x}{4} + C$	Use half-angle: $\cos^2 x = \frac{1+\cos 2x}{2}$

STRATEGY FOR $\int \sin^m x \cos^n x \, dx$

Case 1 — Odd power of $\sin x$ (m odd): Save one $\sin x$, convert the rest using $\sin^2 x = 1 - \cos^2 x$, then u-sub $u = \cos x$.

Case 2 — Odd power of $\cos x$ (n odd): Save one $\cos x$, convert the rest using $\cos^2 x = 1 - \sin^2 x$, then u-sub $u = \sin x$.

Case 3 — Both even: Use half-angle formulas to reduce powers, then integrate.

Case 4 — $\int \tan^m x \sec^n x \, dx$: If n even, save $\sec^2 x$ and express the rest in terms of $\tan x$, then u-sub $u = \tan x$. If m odd, save $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$.

EASY Evaluate $\sin(5\pi/3)$ and $\cos(3\pi/4)$ using CAST and reference angles

$\sin(5\pi/3)$: Q4 (BETWEEN $3\pi/2$ AND 2π), SINE IS NEGATIVE IN Q4

Reference angle: $2\pi - 5\pi/3 = \pi/3$. So $\sin(5\pi/3) = -\sin(\pi/3) = -\frac{\sqrt{3}}{2}$.

$\cos(3\pi/4)$: Q2 (BETWEEN $\pi/2$ AND π), COSINE IS NEGATIVE IN Q2

Reference angle: $\pi - 3\pi/4 = \pi/4$. So $\cos(3\pi/4) = -\cos(\pi/4) = -\frac{\sqrt{2}}{2}$.

MEDIUM Differentiate $f(x) = \sin^3(4x^2)$

TWO APPLICATIONS OF CHAIN RULE: OUTER U^3 , INNER $\sin(U)$, INNERMOST $U=4x^2$

$$f'(x) = 3\sin^2(4x^2) \cdot \frac{d}{dx}[\sin(4x^2)]$$

$$\frac{d}{dx}[\sin(4x^2)] = \cos(4x^2) \cdot 8x$$

$f'(x) = 24x \sin^2(4x^2) \cos(4x^2)$

Note: This can be simplified using the double angle formula: $= 12x \sin(8x^2) \cdot \sin(4x^2) / \sin(4x^2) \dots$ but the form above is exact.

MEDIUM Evaluate $\int \sin^3 x \, dx$

ODD POWER OF SIN: SAVE ONE $\sin x$, CONVERT $\sin^2 x = 1 - \cos^2 x$

$$\int \sin^3 x \, dx = \int \sin x (1 - \cos^2 x) \, dx$$

U-SUB: $U = \cos x \Rightarrow dU = -\sin x dx$

$$= -\int (1 - u^2) \, du = -u + \frac{u^3}{3} + C$$

$\int \sin^3 x \, dx = -\cos x + \frac{\cos^3 x}{3} + C$

HARD Evaluate $\int \cos^4 x \, dx$

BOTH POWERS EVEN. USE HALF-ANGLE: $\cos^2 x = \frac{1 + \cos 2x}{2}$

$$\cos^4 x = (\cos^2 x)^2 = \left(\frac{1 + \cos 2x}{2}\right)^2 = \frac{1 + 2\cos 2x + \cos^2 2x}{4}$$

APPLY HALF-ANGLE AGAIN TO $\cos^2 2x = \frac{1 + \cos 4x}{2}$

$$\cos^4 x = \frac{1}{4} \left(1 + 2\cos 2x + \frac{1 + \cos 4x}{2} \right) = \frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$$

INTEGRATE TERM BY TERM

$\int \cos^4 x \, dx = \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

HARD Verify the identity: $\frac{\tan x + \cot x}{\sec x \csc x} = 1$

CONVERT EVERYTHING TO $\sin$ AND $\cos$

LHS = $\frac{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}{\frac{1}{\cos x} \cdot \frac{1}{\sin x}} = \frac{\frac{\sin^2 x + \cos^2 x}{\sin x \cos x}}{\frac{1}{\sin x \cos x}} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} \cdot \sin x \cos x = 1$

Using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$. ✓

VERY HARD Evaluate $\int \tan^4 x \, dx$

WRITE $\tan^4 x = \tan^2 x \cdot \tan^2 x = \tan^2 x (\sec^2 x - 1)$

$$\int \tan^4 x \, dx = \int \tan^2 x \sec^2 x \, dx - \int \tan^2 x \, dx$$

FIRST INTEGRAL: U-SUB $U = \tan x$

$$\int \tan^2 x \sec^2 x \, dx = \int u^2 \, du = \frac{u^3}{3} = \frac{\tan^3 x}{3}$$

SECOND INTEGRAL: $\tan^2 x = \sec^2 x - 1$

$$\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \tan x - x$$

COMBINE

$$\int \tan^4 x \, dx = \frac{\tan^3 x}{3} - \tan x + x + C$$

⚠ COMMON TRAPS — TRIGONOMETRY

Trap 1 — Derivative of $\cos x$ sign: $\frac{d}{dx}[\cos x] = -\sin x$. The minus sign is frequently dropped.

Trap 2 — Quadrant sign errors: Always apply CAST first before evaluating trig at angles beyond $\pi/2$. For example, $\sin(5\pi/6) = +\frac{1}{2}$ (Q2, sin positive), NOT $-\frac{1}{2}$.

Trap 3 — $\sin^2 x$ vs $\sin(x^2)$: These are completely different. $\sin^2 x = (\sin x)^2$; use chain rule for $\sin(x^2)$.

Trap 4 — $\int \sin x \, dx = +\cos x$ (wrong sign): The antiderivative of $\sin x$ is $-\cos x$, NOT $+\cos x$.

Trap 5 — Wrong Pythagorean identity form: $1 + \tan^2 x = \sec^2 x$, so $\tan^2 x = \sec^2 x - 1$. Don't confuse this with $\sec^2 x - \tan^2 x = 1$ (both are correct rearrangements).

Trap 6 — Forgetting domains of inverse trig: $\arcsin$ and $\arccos$ only accept inputs in $[-1, 1]$. $\arctan$ accepts all reals. Don't try to evaluate $\arcsin(2)$.

§ 06 Implicit Differentiation

Differentiating equations that cannot (or need not) be solved for y explicitly — with full technique and exhaustive examples.

6.1 — DEFINITION & METHOD

IMPLICIT DIFFERENTIATION — THE CORE IDEA

When a relationship between x and y is given implicitly (e.g. $x^2 + y^2 = r^2$), we differentiate both sides with respect to x , treating y as an implicit function of x . Every time we differentiate a term involving y , we apply the Chain Rule and multiply by $\frac{dy}{dx}$.

General principle:

$$\frac{d}{dx}[f(y)] = f'(y) \cdot \frac{dy}{dx}$$

For example:

$$\frac{d}{dx}[y^3] = 3y^2 \frac{dy}{dx}, \quad \frac{d}{dx}[\sin y] = \cos y \frac{dy}{dx}$$

After differentiating, collect all $\frac{dy}{dx}$ terms on one side and solve algebraically.

METHOD – STEP BY STEP

- 1. Differentiate both sides of the equation with respect to x .
- 2. Every y -term picks up a $\frac{dy}{dx}$ factor by the Chain Rule.
- 3. For mixed terms (like xy), apply the Product Rule: $\frac{d}{dx}[xy] = y + x\frac{dy}{dx}$.
- 4. Collect all $\frac{dy}{dx}$ terms on the left side.
- 5. Factor out $\frac{dy}{dx}$ and solve.
- 6. If finding the second derivative $\frac{d^2y}{dx^2}$, differentiate the expression for $\frac{dy}{dx}$ again, substituting the expression for $\frac{dy}{dx}$ as needed.

6.2 — WORKED EXAMPLES

EASY Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$ (circle of radius 5)

DIFFERENTIATE BOTH SIDES W.R.T. x

$$2x + 2y \frac{dy}{dx} = 0$$

SOLVE FOR $\frac{dy}{dx}$

$$\frac{dy}{dx} = -\frac{x}{y}$$

This represents the slope of the tangent to the circle at any point (x, y) . At $(3, 4)$: $\frac{dy}{dx} = -\frac{3}{4}$.

MEDIUM Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$ (Folium of Descartes)

DIFFERENTIATE BOTH SIDES – NOTE THE PRODUCT RULE ON $6xy$

$$3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}$$

COLLECT $\frac{dy}{dx}$ TERMS ON THE LEFT

$$3y^2 \frac{dy}{dx} - 6x \frac{dy}{dx} = 6y - 3x^2$$

$$\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$$

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$$

MEDIUM Find $\frac{dy}{dx}$ for $e^{xy} + \ln(y^2) = x^3$

DIFFERENTIATE LHS TERM 1: e^{xy} . CHAIN RULE – INNER IS xy , PRODUCT RULE GIVES $(Y + X\frac{dy}{dx})$

$$\frac{d}{dx}[e^{xy}] = e^{xy}\left(y + x\frac{dy}{dx}\right)$$

DIFFERENTIATE LHS TERM 2: $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$

$$\frac{d}{dx}[2\ln y] = \frac{2}{y}\frac{dy}{dx}$$

RHS: $\frac{d}{dx}[x^3] = 3x^2$. ASSEMBLE AND SOLVE FOR $\frac{dy}{dx}$

$$ye^{xy} + xe^{xy}\frac{dy}{dx} + \frac{2}{y}\frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx}\left(xe^{xy} + \frac{2}{y}\right) = 3x^2 - ye^{xy}$$

$$\frac{dy}{dx} = \frac{3x^2 - ye^{xy}}{xe^{xy} + \frac{2}{y}}$$

HARD Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = r^2$

FROM THE PREVIOUS EASY EXAMPLE: $\frac{dy}{dx} = -\frac{x}{y}$

DIFFERENTIATE $\frac{dy}{dx}$ W.R.T. x USING QUOTIENT RULE

$$\frac{d^2y}{dx^2} = -\frac{y \cdot 1 - x \cdot \frac{dy}{dx}}{y^2} = -\frac{y - x \cdot \left(-\frac{x}{y}\right)}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2}$$

SIMPLIFY

$$\frac{d^2y}{dx^2} = -\frac{\frac{y^2+x^2}{y}}{y^2} = -\frac{x^2 + y^2}{y^3}$$

Since $x^2 + y^2 = r^2$:

$$\frac{d^2y}{dx^2} = -\frac{r^2}{y^3}$$

VERY HARD Find $\frac{dy}{dx}$ for $\sin(xy) + x^2y^3 = \cos x + y^4$

DIFFERENTIATE EACH TERM

$$\frac{d}{dx}[\sin(xy)] = \cos(xy)\left(y + x\frac{dy}{dx}\right)$$

$$\frac{d}{dx}[x^2y^3] = 2xy^3 + 3x^2y^2\frac{dy}{dx}$$

$$\frac{d}{dx}[\cos x] = -\sin x$$

$$\frac{d}{dx}[y^4] = 4y^3\frac{dy}{dx}$$

ASSEMBLE THE FULL EQUATION

$$y\cos(xy) + x\cos(xy)\frac{dy}{dx} + 2xy^3 + 3x^2y^2\frac{dy}{dx} = -\sin x + 4y^3\frac{dy}{dx}$$

COLLECT ALL $\frac{Dy}{Dx}$ TERMS

$$\frac{dy}{dx}[x\cos(xy) + 3x^2y^2 - 4y^3] = -\sin x - y\cos(xy) - 2xy^3$$

$$\frac{dy}{dx} = \frac{-\sin x - y\cos(xy) - 2xy^3}{x\cos(xy) + 3x^2y^2 - 4y^3}$$

⚠ COMMON TRAPS – IMPLICIT DIFFERENTIATION

Trap 1 — Forgetting $\frac{dy}{dx}$ on y -terms: $\frac{d}{dx}[y^2] = 2y\frac{dy}{dx}$, NOT just $2y$. The Chain Rule is mandatory since $y = y(x)$.

Trap 2 — Product Rule on mixed terms: $\frac{d}{dx}[xy^2] = y^2 + 2xy\frac{dy}{dx}$. Both factors are functions of x , so the Product Rule is required.

Trap 3 — Moving all $\frac{dy}{dx}$ to the same side: Before factoring, make sure every term containing $\frac{dy}{dx}$ is on the left, everything else on the right.

Trap 4 — Second derivative substitution: When computing $\frac{d^2y}{dx^2}$, after differentiating once more, substitute the expression for $\frac{dy}{dx}$ back in — don't leave the result with $\frac{dy}{dx}$ in it unless y is also present.

S 07 **Curve Sketching with Derivatives**

Using the first and second derivative to classify critical points, determine concavity, find inflection points, and produce accurate curve sketches.

7.1 — THE SYSTEMATIC METHOD

COMPLETE CURVE SKETCHING CHECKLIST

- 1. **Domain:** Find all x values where $f(x)$ is defined.
- 2. **Intercepts:** x -intercepts: set $f(x) = 0$. y -intercept: evaluate $f(0)$.
- 3. **Symmetry:** Even ($f(-x) = f(x)$), odd ($f(-x) = -f(x)$), or neither.
- 4. **Asymptotes:** Vertical (where $f \rightarrow \pm\infty$), horizontal ($\lim_{x \rightarrow \pm\infty} f(x)$), oblique (polynomial long division if applicable).
- 5. **First Derivative $f'(x)$:** Find and solve $f'(x) = 0$ and where $f'(x)$ is undefined. These are critical points.
- 6. **Increasing / Decreasing:** Sign of $f'(x)$ on each interval: + means increasing, – means decreasing.
- 7. **Local Extrema:** Apply First or Second Derivative Test at critical points.

8. **Second Derivative** $f''(x)$: Find and solve $f''(x) = 0$. These are potential inflection points.
9. **Concavity**: $f'' > 0$ means concave up (smile); $f'' < 0$ means concave down (frown).
10. **Inflection Points**: Where concavity changes and $f''(x) = 0$ (or undefined).
11. **Sketch**: Plot key points and connect smoothly, respecting all the above information.

7.2 — CRITICAL POINTS & THE DERIVATIVE TESTS

FIRST DERIVATIVE TEST

At a critical point $x = c$ where $f'(c) = 0$:

f' changes $- \rightarrow +$: local **minimum**

f' changes $+ \rightarrow -$: local **maximum**

f' does not change sign: **saddle point** (neither)

SECOND DERIVATIVE TEST

At $x = c$ where $f'(c) = 0$:

$f''(c) > 0$: local **minimum**

$f''(c) < 0$: local **maximum**

$f''(c) = 0$: test is **inconclusive** — use the first derivative test

7.3 — CONCAVITY & INFLECTION POINTS

CONCAVITY — FORMAL DEFINITION

A function is **concave up** on an interval if $f''(x) > 0$ throughout — the graph lies above its tangent lines (holds water). It is **concave down** if $f''(x) < 0$ — the graph lies below its tangent lines (sheds water).

An **inflection point** is a point where the concavity changes. It requires $f''(c) = 0$ (or $f''(c)$ undefined) AND a sign change in f'' at c . Merely having $f''(c) = 0$ is necessary but not sufficient.

Key relationship: Where $f''(x) > 0$, the first derivative $f'(x)$ is increasing (the function is accelerating). Where $f''(x) < 0$, $f'(x)$ is decreasing.

7.4 — WORKED EXAMPLES

MEDIUM Sketch $f(x) = x^3 - 3x^2 + 2$

DOMAIN & INTERCEPTS

Domain: all reals. $f(0) = 2$. For roots: $x^3 - 3x^2 + 2 = 0 \Rightarrow (x - 1)(x^2 - 2x - 2) = 0$ giving $x = 1$ and $x = 1 \pm \sqrt{3}$.

FIRST DERIVATIVE

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

Critical points: $x = 0$ and $x = 2$.

SIGN CHART FOR $f'(x)$

INTERVAL	$f'(x)$	f
$(-\infty, 0)$	+	Increasing
$(0, 2)$	−	Decreasing
$(2, \infty)$	+	Increasing

$\Rightarrow$ Local max at $x = 0$: $f(0) = 2$. Local min at $x = 2$: $f(2) = 8 - 12 + 2 = -2$.

SECOND DERIVATIVE

$$f''(x) = 6x - 6$$

$f''(x) = 0 \Rightarrow x = 1$. $f''(x) < 0$ on $(-\infty, 1)$ (concave down); $f''(x) > 0$ on $(1, \infty)$ (concave up). Inflection point at $(1, f(1)) = (1, 0)$.

SECOND DERIVATIVE TEST AT CRITICAL POINTS

$f''(0) = -6 < 0 \Rightarrow$ local max confirmed. $f''(2) = 6 > 0 \Rightarrow$ local min confirmed. **Summary:** Rises from $-\infty$, reaches local max at $(0, 2)$, falls to local min at $(2, -2)$, rises to $+\infty$. Inflection at $(1, 0)$.

HARD Sketch $f(x) = \frac{x^2}{x^2 - 4}$ (includes asymptotes)

DOMAIN

$x^2 - 4 \neq 0 \Rightarrow x \neq \pm 2$. Domain: $\mathbb{R} \setminus \{-2, 2\}$.

INTERCEPTS

$f(0) = 0$ (y-intercept). x -intercept: $x^2 = 0 \Rightarrow x = 0$. Only one intercept at the origin.

ASYMPTOTES

Vertical: $x = \pm 2$ (denominator zero). Horizontal: $\lim_{x \rightarrow \pm\infty} \frac{x^2}{x^2 - 4} = 1$, so $y = 1$. Note: $f(x) = 1 + \frac{4}{x^2 - 4}$, so $f(x) > 1$ when $x^2 > 4$ and $f(x) < 1$ when $|x| < 2$.

FIRST DERIVATIVE (QUOTIENT RULE)

$$f'(x) = \frac{2x(x^2 - 4) - x^2 \cdot 2x}{(x^2 - 4)^2} = \frac{2x^3 - 8x - 2x^3}{(x^2 - 4)^2} = \frac{-8x}{(x^2 - 4)^2}$$

Critical point: $f'(x) = 0 \Rightarrow x = 0$. $f'(x) > 0$ for $x < 0$, $f'(x) < 0$ for $x > 0$ (where defined). So local max at $x = 0$: $f(0) = 0$.

SECOND DERIVATIVE (TO CONFIRM)

$$f''(x) = \frac{-8(x^2 - 4)^2 - (-8x) \cdot 2(x^2 - 4) \cdot 2x}{(x^2 - 4)^4} = \frac{-8(x^2 - 4) + 32x^2}{(x^2 - 4)^3} = \frac{24x^2 + 32}{(x^2 - 4)^3}$$

$$f''(0) = \frac{32}{(-4)^3} = \frac{32}{-64} = -\frac{1}{2} < 0. \text{ Confirms local max.}$$

SUMMARY

Even function (symmetric about y -axis). Approaches $y = 1$ from below for $|x| < 2$ and from above for $|x| > 2$. Vertical asymptotes at $x = \pm 2$. Local max at origin.

VERY HARD Sketch $f(x) = xe^{-x}$ — full analysis

DOMAIN & INTERCEPTS

All reals. $f(0) = 0$. $f(x) = 0 \Rightarrow x = 0$ only (since $e^{-x} > 0$ always).

ASYMPTOTES

As $x \rightarrow +\infty$: $xe^{-x} \rightarrow 0$ (exponential decay beats polynomial growth). As $x \rightarrow -\infty$: $xe^{-x} \rightarrow -\infty$ (both factors grow). Horizontal asymptote: $y = 0$ on the right only.

FIRST DERIVATIVE (PRODUCT RULE)

$$f'(x) = e^{-x} + x(-e^{-x}) = e^{-x}(1 - x)$$

Critical point: $f'(x) = 0 \Rightarrow 1 - x = 0 \Rightarrow x = 1$. Since $e^{-x} > 0$ always: $f' > 0$ for $x < 1$ (increasing), $f' < 0$ for $x > 1$ (decreasing). Local max at $x = 1$: $f(1) = e^{-1} = 1/e$.

SECOND DERIVATIVE

$$f''(x) = -e^{-x}(1 - x) + e^{-x}(-1) = e^{-x}(x - 2)$$

$f''(x) = 0 \Rightarrow x = 2$. Sign: $f'' < 0$ for $x < 2$ (concave down), $f'' > 0$ for $x > 2$ (concave up). Inflection point at $(2, 2e^{-2})$.

SUMMARY OF KEY POINTS

Origin, local max at $(1, \frac{1}{e})$, inflection at $(2, \frac{2}{e^2})$, horizontal asymptote $y = 0$ as $x \rightarrow +\infty$.

⚠ COMMON TRAPS — CURVE SKETCHING

Trap 1 — $f''(c) = 0$ does NOT guarantee an inflection point: For $f(x) = x^4$, $f''(0) = 0$ but the function is concave up on both sides — no inflection occurs. You must verify a sign change in f'' .

Trap 2 — Inconclusive Second Derivative Test: When $f''(c) = 0$, switch to the First Derivative Test instead.

Trap 3 — Missing vertical asymptotes in the critical point search: Points where $f'(x)$ is undefined (due to vertical asymptotes) are also critical numbers and must be included in the sign chart.

Trap 4 — Horizontal asymptote sign: The function can cross its horizontal asymptote for finite x — the asymptote only describes behaviour as $x \rightarrow \pm\infty$.

§ 08

Volumes of Revolution

Computing the volume generated when a region is rotated about an axis — using the Disk, Washer, and Shell methods.

8.1 — DISK & WASHER METHOD

DISK METHOD — ROTATION ABOUT THE x -AXIS

When the region under $y = f(x) \geq 0$ on $[a, b]$ is rotated about the x -axis, each cross-section is a disk of radius $f(x)$:

$$V = \pi \int_a^b [f(x)]^2 dx$$

Intuition: $\pi r^2 \cdot dx$ — the volume of an infinitely thin circular slice of radius $r = f(x)$ and thickness dx .

WASHER METHOD — ROTATION ABOUT THE x -AXIS (REGION BETWEEN TWO CURVES)

When the region between $y = f(x)$ (outer) and $y = g(x)$ (inner), with $f(x) \geq g(x) \geq 0$, is rotated about the x -axis, each cross-section is a washer (annulus):

$$V = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx$$

ROTATION ABOUT THE y -AXIS (DISK/WASHER)

Express x as a function of y : $x = h(y)$. Integrate with respect to y from c to d :

$$V = \pi \int_c^d [h(y)]^2 dy$$

For a washer: $V = \pi \int_c^d ([h_{\text{outer}}(y)]^2 - [h_{\text{inner}}(y)]^2) dy$

8.2 — SHELL METHOD (CYLINDRICAL SHELLS)

SHELL METHOD — ROTATION ABOUT THE y -AXIS

When the region under $y = f(x)$, $f(x) \geq 0$, on $[a, b]$ (with $a \geq 0$) is rotated about the y -axis:

$$V = 2\pi \int_a^b x f(x) dx$$

Intuition: Each vertical strip of width dx at position x sweeps out a thin cylindrical shell of circumference $2\pi x$, height $f(x)$, and thickness dx . Volume of shell $\approx 2\pi x \cdot f(x) \cdot dx$.

Shell method about the x -axis:

$$V = 2\pi \int_c^d y h(y) dy$$

where $x = h(y)$ and the region is integrated along y .

CHOOSING THE RIGHT METHOD

Disk/Washer: Use when cross-sections perpendicular to the axis of rotation are easy to describe. The integration variable is parallel to the axis of rotation.

Shell: Use when the cross-sections perpendicular to the axis are hard (would require expressing x as a function of y , say), or when the region is described most naturally in the other variable. Integration is parallel to the axis of rotation's perpendicular.

Rule of thumb: Rotating about the x -axis $\rightarrow$ use disks/washers in x . Rotating about the y -axis $\rightarrow$ shells in x are usually easier (avoids inverting the function).

DISK (X-AXIS)

$$V = \pi \int_a^b [f(x)]^2 dx$$

WASHER (X-AXIS)

$$V = \pi \int_a^b ([f]^2 - [g]^2) dx$$

SHELL (Y-AXIS)

$$V = 2\pi \int_a^b x f(x) dx$$

SHELL (X-AXIS)

$$V = 2\pi \int_c^d y h(y) dy$$

8.3 — WORKED EXAMPLES

EASY Volume of a sphere of radius r using the disk method

ROTATE $Y = \sqrt{R^2 - X^2}$ ABOUT THE X -AXIS ON $[-R, R]$

$$V = \pi \int_{-r}^r (r^2 - x^2) dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r$$

At $x = r$: $r^3 - r^3/3 = 2r^3/3$. At $x = -r$: $-r^3 + r^3/3 = -2r^3/3$.

$$V = \pi \left(\frac{2r^3}{3} + \frac{2r^3}{3} \right) = \boxed{\frac{4}{3} \pi r^3}$$

MEDIUM Volume obtained by rotating the region bounded by $y = \sqrt{x}$, $y = 0$, $x = 4$ about the x -axis

USE THE DISK METHOD

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi \cdot 8 = \boxed{8\pi}$$

MEDIUM Volume using Washer Method: region between $y = x^2$ and $y = x$ rotated about the x -axis

INTERSECTION: $X^2 = X \Rightarrow X = 0, 1$. ON $[0, 1]$: $X \geq X^2$. OUTER RADIUS $R = X$, INNER RADIUS $R = X^2$

$$\begin{aligned} V &= \pi \int_0^1 (x^2 - (x^2)^2) dx = \pi \int_0^1 (x^2 - x^4) dx \\ &= \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \boxed{\frac{2\pi}{15}} \end{aligned}$$

HARD Volume using Shell Method: region $y = e^{-x^2}$, $x = 0$, $x = 1$, $y = 0$ rotated about the y -axis

SHELL METHOD: $V = 2\pi \int_0^1 X \cdot E^{X^2} DX$

U-SUB: $U = X^2 \Rightarrow DU = 2XDX \Rightarrow XDX = DU/2$. BOUNDS: $0 \rightarrow 0, 1 \rightarrow 1$

$$\begin{aligned} V &= 2\pi \int_0^1 e^{-u} \cdot \frac{du}{2} = \pi \int_0^1 e^{-u} du = \pi [-e^{-u}]_0^1 = \pi(1 - e^{-1}) \\ &= \boxed{\pi \left(1 - \frac{1}{e} \right)} \end{aligned}$$

Note: Attempting the disk method here would require $x = \sqrt{-\ln y}$, leading to a much harder integral. The shell method is clearly superior.

VERY HARD Volume between $y = \sin x$ and $y = \cos x$ on $[0, \pi/4]$, rotated about $y = -1$

THE AXIS OF ROTATION IS $Y = -1$, NOT THE X -AXIS. SHIFT RADII ACCORDINGLY

On $[0, \pi/4]$: $\cos x \geq \sin x$. Outer function from axis $y = -1$: $R = \cos x - (-1) = \cos x + 1$. Inner function: $r = \sin x + 1$.

WASHER METHOD WITH SHIFTED AXIS

$$\begin{aligned} V &= \pi \int_0^{\pi/4} [(\cos x + 1)^2 - (\sin x + 1)^2] dx \\ &= \pi \int_0^{\pi/4} [\cos^2 x + 2\cos x + 1 - \sin^2 x - 2\sin x - 1] dx \\ &= \pi \int_0^{\pi/4} [\cos 2x + 2(\cos x - \sin x)] dx \end{aligned}$$

(using $\cos^2 x - \sin^2 x = \cos 2x$)

$$= \pi \left[\frac{\sin 2x}{2} + 2(\sin x + \cos x) \right]_0^{\pi/4}$$

At $x = \pi/4$: $\frac{\sin(\pi/2)}{2} + 2\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) = \frac{1}{2} + 2\sqrt{2}$. At $x = 0$: $0 + 2(0 + 1) = 2$.

$$V = \pi \left(\frac{1}{2} + 2\sqrt{2} - 2 \right) = \pi \left(2\sqrt{2} - \frac{3}{2} \right)$$

⚠ COMMON TRAPS – VOLUMES OF REVOLUTION

Trap 1 — Squaring the whole integral, not the function: $V = \pi \int_a^b [f(x)]^2 dx$, NOT $\pi \left(\int_a^b f(x) dx \right)^2$. The square goes inside the integral.

Trap 2 — Washer with wrong radii: The outer radius is the function farther from the axis of rotation. If rotating about $y = -2$, the radii are $f(x) + 2$ and $g(x) + 2$, not $f(x)$ and $g(x)$.

Trap 3 — Shell method sign: The shell formula $V = 2\pi \int_a^b x f(x) dx$ requires $a \geq 0$. For regions that cross the axis of rotation, split the integral.

Trap 4 — Choosing the wrong method: Both methods must give the same answer. If one method leads to an impossible or very difficult integral (like inverting $y = e^{-x^2}$), switch methods.

